

Overview of Denoising diffusion probabilistic model

Subject: Denoising diffusion probabilistic model

1.

$$x_t | x_0 \sim N\left(x_0 - \frac{1}{2} \int_0^t \beta(s) ds, \left(1 - e^{-\int_0^t \beta(s) ds}\right) I\right)$$

2.

Sample the previous data $x_{t-1} \sim N(\mu_{\theta}(x_t, x_0), \sigma_{\theta}^2 I)$

3.

Noise prediction network Since $x_{t-1} | x_t, x_0 \sim N(\mu_{\theta}(x_t, x_0), \sigma_{\theta}^2 I)$, this suggests that we should use $\mu_{\theta}(x_t, t) = \mu_{\theta}(x_t, x_0)$; however, the network does not have access to x_0 , and so it has to estimate it instead. Now, since $x_t | x_0 \sim N\left(x_0 - \frac{1}{2} \int_0^t \beta(s) ds, \left(1 - e^{-\int_0^t \beta(s) ds}\right) I\right)$, we may write $x_t = x_0 + \sigma_t z$, where z is some unknown Gaussian noise. Now we see that estimating x_0 is equivalent to estimating z . Therefore, let the network output a noise vector $\epsilon_{\theta}(x_t, t)$, and let it predict $\mu_{\theta}(x_t, t) = \mu_{\theta}(x_t, x_t - \sigma_t \epsilon_{\theta}(x_t, t) / \sigma_t) = \mu_{\theta}(x_t, t) \beta_t / \sigma_t$. It remains to design $\Sigma_{\theta}(x_t, t)$. The DDPM paper suggested not learning it (since it resulted in "unstable training and poorer sample quality"), but fixing it at some value $\Sigma_{\theta}(x_t, t) = \zeta_t^2 I$, where either $\zeta_t^2 = \beta_t$ or $\zeta_t^2 = \sigma_t^2$ yielded similar performance. With this, the loss simplifies to $L_t = \frac{\beta_t}{2\alpha_t \sigma_t \zeta_t^2} E_{x_0 \sim q; z \sim N(0, I)} [\|\epsilon_{\theta}(x_t, t) - z\|^2] + C$ which may be minimized by stochastic gradient descent. The paper noted empirically that an even simpler loss function $L_{\text{simple}, t} = E_{x_0 \sim q; z \sim N(0, I)} [\|\epsilon_{\theta}(x_t, t) - z\|^2]$ resulted in better models.

4.

One, we no longer need to normalize $q(x)$, but can use any $q \sim (x) = C q(x)$, where $C = \int q \sim (x) dx > 0$ is any unknown constant that is of no concern to us. Two, we are comparing $q(x)$

$\{q(x)\}$ neighbors $q(x+dx)$, by $q(x)q(x+dx) = e^{-\nabla_x \ln q, dx}$

5.

Their equivalence DDPM and score-based generative models are equivalent. This means that a network trained using DDPM can be used as a NCSN, and vice versa. We know that $x_t | x_0 \sim N(\alpha_t x_0, \sigma_t^2 I)$, so by Tweedie's formula, we have

6.

Sample (x_0, z_0, c) , where x_0 is the high-resolution image, z_0 is the same image but scaled down to a low-resolution, and c is the conditioning, which can be the caption of the image, the class of the image, etc. Sample two white noises ϵ_x, ϵ_z , two time-steps t_x, t_z . Compute the noisy versions of the high-resolution and low-resolution images: $x_{t_x} = \alpha_{t_x} x_0 + \sigma_{t_x} \epsilon_x$, $z_{t_z} = \alpha_{t_z} z_0 + \sigma_{t_z} \epsilon_z$. Train the denoising network to predict ϵ_x given $x_{t_x}, z_{t_z}, t_x, t_z, c$. That is, apply gradient descent on θ on the L2 loss $\|\epsilon_\theta(x_{t_x}, z_{t_z}, t_x, t_z, c) - \epsilon_x\|_2^2$.

7.

Simple, $t = E_{x_0, x_t \sim q} [\|\epsilon_\theta(x_t, t) - x_t - \alpha_t^{-1} x_0 \sigma_t\|^2] = E_{x_t \sim q, x_0 \sim q(\cdot | x_t)} [\|\epsilon_\theta(x_t, t) - x_t - \alpha_t^{-1} x_0 \sigma_t\|^2] = E_{x_t \sim q, x_0 \sim q(\cdot | x_t)} [\|\epsilon_\theta(x_t, t) - x_t - \alpha_t^{-1} x_0 \sigma_t\|^2]$